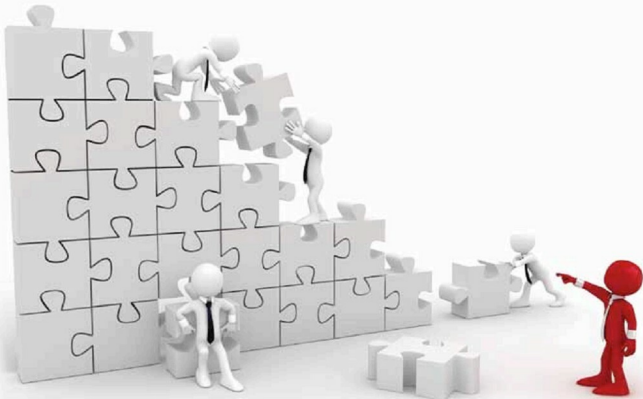


Connect All Data Sources Effectively

Very often a situation may arise when it is necessary to display information that is stored within various sources - databases, Web pages, etc. In many cases, these data sources could have one common feature—for example, they could share a unique identifier. It's possible to prepare and combine these data into one in a rather simple manner – by combining shell-code spiced with regular expressions and *sed*, as well as JDBC-driver and simple JSP-code.



Let's begin by imagining that there are several warehouses, with each storing the same catalogue of products (every warehouse will later on be represented as PostgreSQL-base). All warehouses are organised into a hierarchy—Base1, Base2 and Base3. In our example, Base1 is a child object to Base2, and Base2 is a child object to Base3. These products steadily migrate from the youngest warehouse (database) to its parent; so we have one source for accumulated information at the end—Base3. How products are transferred from one warehouse to another is not important—we just need to have the information on every object that physically moves, and store this information in its particular database. The migration process from Base1 to Base2, and from Base2 to Base3 is essentially the same. It is important to note that the root database (Base3) accumulates information

not from just a single offspring Base1/ Base2, but also gets data from a number of neighbouring companies. A prime example for such a structure could be a national retail chain, which has its headquarters in the capital city, and has a number of branches and offices across the country. The database stores information about all products sold throughout the distribution network. Or this could be any other information that has a tendency to get accumulated in a single unified database being deployed at headquarters. A real-world example for this distributed database could be the common snail-mail network for a national post-office service.

Everyday activity requires one to perform an audit of the status of goods (or messages)—whether delivered from the branch to headquarters or not; and if delivered, when? Quite obviously, this verification process has a cycle, which is